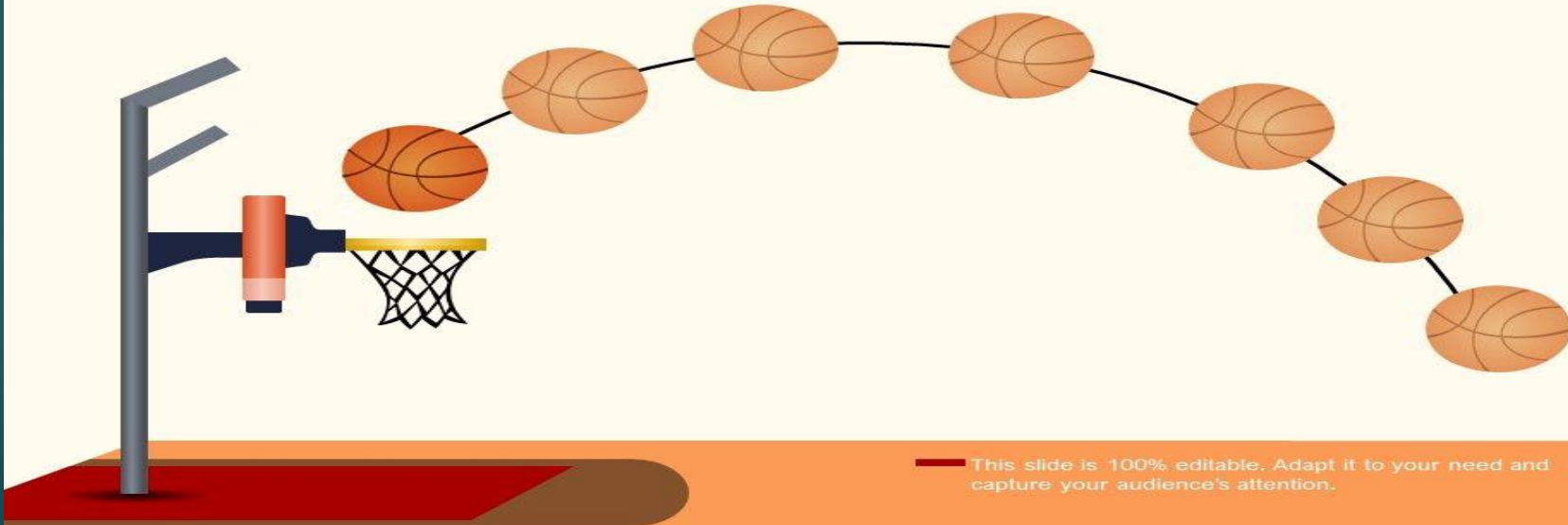


Experiment #(2)

Projectile Motion

(حركة المقذوفات)

Projectile Motion Parabolic with Basket Ball



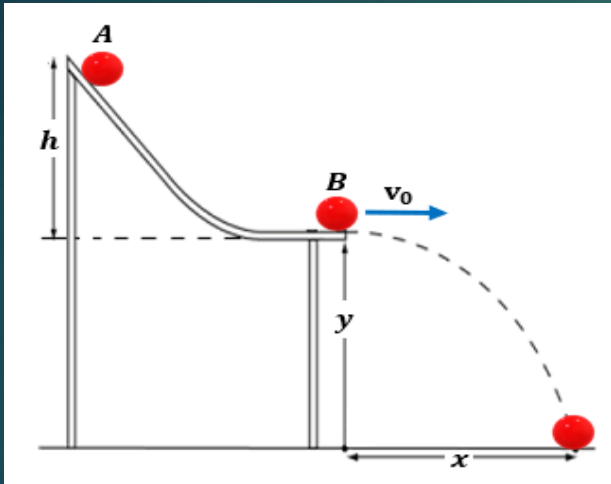
This slide is 100% editable. Adapt it to your need and capture your audience's attention.

Introduction

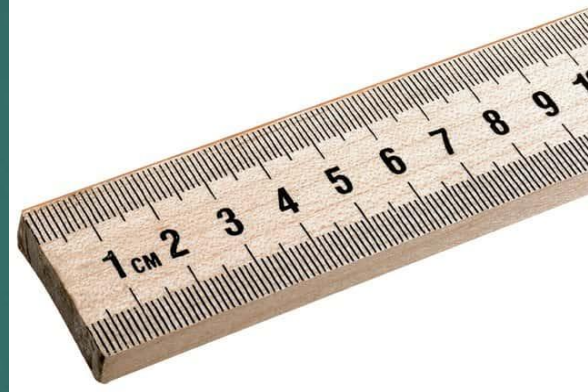
The projectile motion is an example of the motion in two dimensions under constant acceleration. The projectile is an object thrown into the air and its motion is only influenced by the constant gravitational acceleration. In this experiment, we use the equations of motion of a projectile in two dimensions to determine the earth's gravitational acceleration.

Equipment's

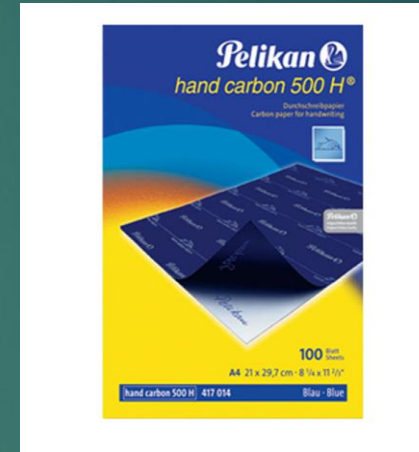
Projectile track



meter stick



carbon papers



Glass Balls



Papers



Theory

The following graph shows a projectile motion of an object from point B to point C as shown with angle $\theta = 0^\circ$ then

$$V_{0x} = V_0 \cos 0 = V_0$$

$$V_{0y} = V_0 \sin 0 = 0$$

The equations of motion along the x-axis are.

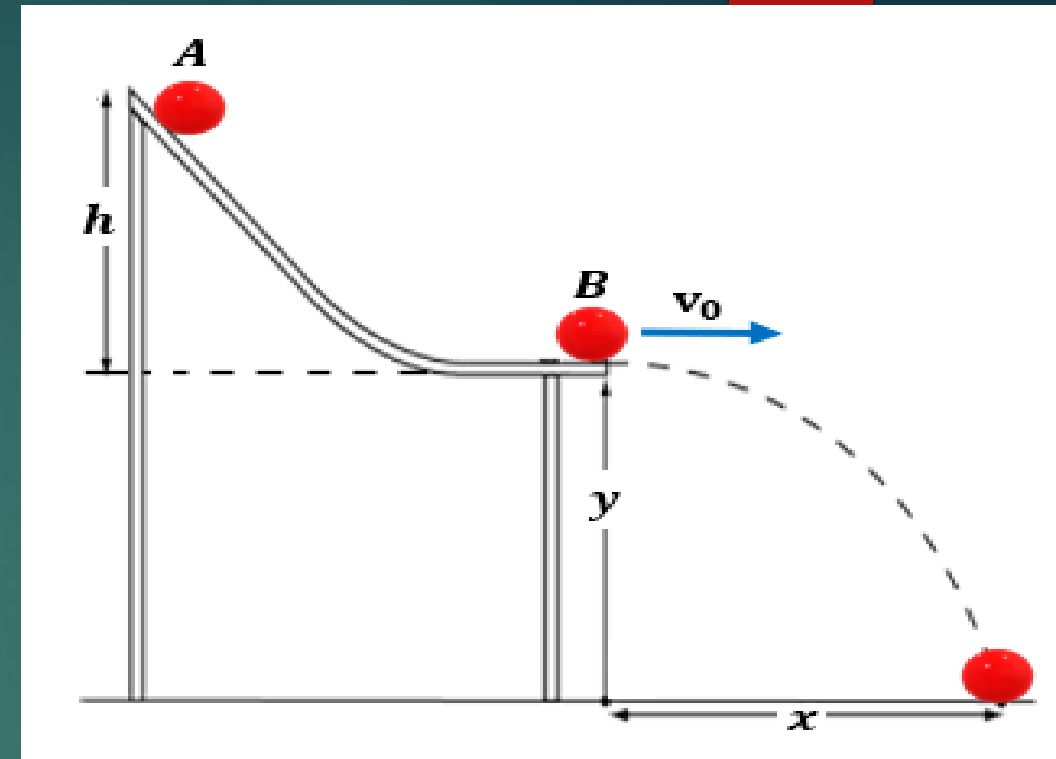
$$v_{xf} = v_{xi} \quad x_f - x_i = v_{xi} t$$

The equations of motion along the y-axis are.

$$v_{yf} = v_{yi} - g t \quad y_f - y_i = v_{yi} t - \frac{1}{2} g t^2$$

Substitute for $x_i = 0$, $x_f = x$ and $v_{xi} = v_0$ in the equations of motion along the x-axis and for $y_i = 0$, $y_f = -y$ and $v_{yi} = 0$ in the equations of motion along the y-axis. Solve the equations to represent y in terms of x

$$y = g \frac{x^2}{2v_0^2}$$



From the conservation of energy from A to B

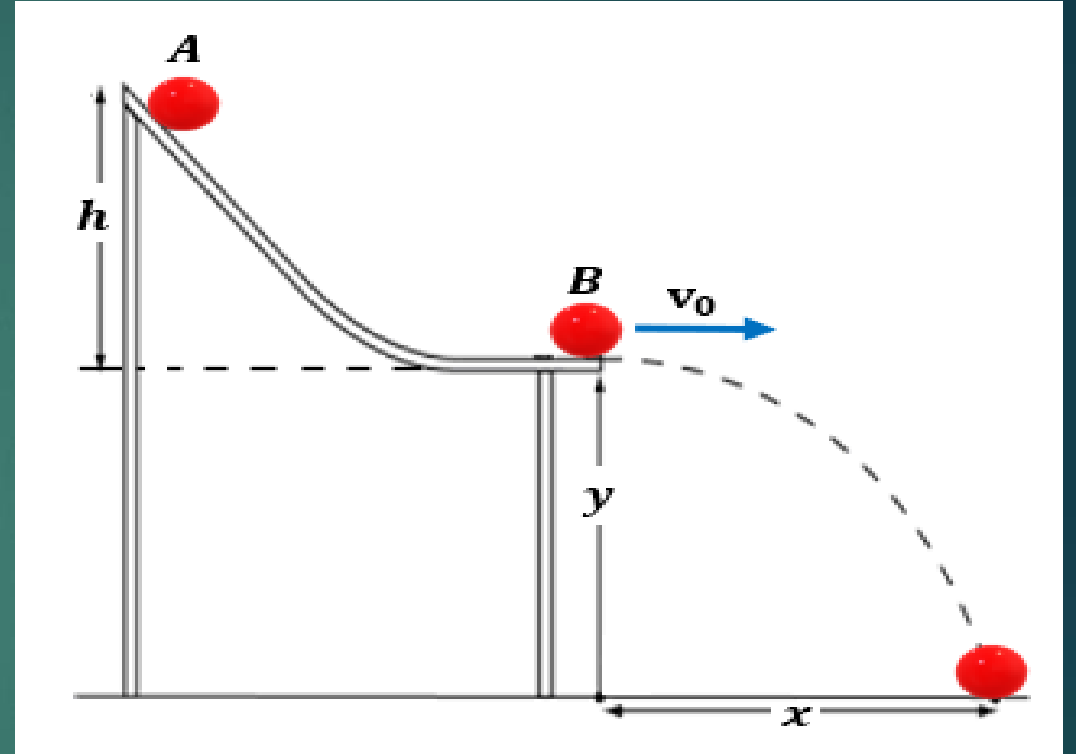
$$E_A = E_B$$

$$mgh = \frac{1}{2} mV_0^2 + \frac{1}{2} I\omega^2$$

Where I : The moment of inertia of the sphere ($I = \frac{2}{5}mr^2$), ω is the angular speed $\omega = \frac{V_0}{r}$

$$V_0 = \sqrt{\frac{10}{7}gh}$$

If $h = \text{constant}$. Then $V_0 = \text{constant}$



Procedure

- Release the ball from point A at height ($h = 15\text{cm}$).
- Calculate the initial velocity using **Equation 2**
- Measure the vertical distance y and the horizontal distance x .
- Use the same height (h), change the vertical distance y five times and measure the corresponding horizontal distance x .
- Record your results in table (1).
- Plot y versus x^2 on a graph paper and draw the best straight line through the points.
- Find the slope, and from the slope calculate the earth's gravitational acceleration.
- Calculate the absolute and relative errors in the earth's gravitational acceleration.
- What are the possible sources of errors in this experiment?

$h = 15\text{ cm}$		$v_0 =$ ()	
Trial	y () $\pm$ ()	x () $\pm$ ()	x^2 ()
1			
2			
3			
4			
5			

THANK YOU
